

in standard alpha calculations (see chapter 10), so often it appears that HFs make money relative to long-only positions in plain-vanilla fixed income and equities.

Many HF strategies have short put payoffs.³⁵ Let's take the HF styles in Figure 17.6:

Long-Short Equity

A common quant HF strategy is short-term reversal, which takes advantage of the fact that if a stock has gone up in price today, it is likely to mean-revert back down tomorrow.³⁶ (This effect is observed in horizons less than a minute to one month.) The predictability of this pattern is weak (like all predictability in stock returns, see chapter 8), but quants use this strategy on thousands of stocks.³⁷ Lo (2007) reports that returns to this strategy have been declining over time. To maintain the overall average of their returns, quant funds juiced up this strategy with more leverage through the 2000s. This is partly what made the August 2007 losses so severe.

Nagel (2012) argues that short-term reversal strategies are a form of liquidity provision—prices decline because other market participants wish to sell, and quant HFs provide liquidity by picking up the slack. Liquidity provision is a short put—you make profits most of the time, but you take a drubbing when prices fall. The profitability of liquidity provision strategies, including short-term reversal, is highly dependent on volatility risk. As volatility jumps, liquidity evaporates merger, saddling long-short equity HFs with losses.

Merger Arb

Merger arbitrage (*merger arb* or *risk arb*) is a strategy that buys stock in companies being acquired (and shorts the acquirer). When a merger is announced, prices of the company being acquired jump, but they don't jump all the way up to the offer price. They jump on the announcement and then continue to drift upward, reaching the offer price when the merger is completed.

Mitchell and Pulvino (2001) show that merger arb is equivalent to a naked short put. In most cases, merger arb makes money because the stock of the acquired company rises to the offer price predictably. The reason prices don't jump upward immediately on announcement of the merger is that sometimes mergers

³⁵ Some HFs are put buyers, generating small losses most of the time but making a killing when markets tank. These funds lose money in the long run because you need to short volatility to earn the volatility risk premium. Nassim Taleb, the well-known pessimistic forecaster, is associated with one such fund, Universa Investments.

³⁶ This is different from momentum, which refers to the tendency of groups of many stocks with past high returns to exhibit high future returns compared to stocks with past low returns. See chapter 7.

³⁷ In terms of Grinold's (1989) "fundamental law" of active management (see chapter 10), quants take lots of small bets with positive but small chances of making money, rather than making few bets that have a relatively high chance of succeeding.